

COMPLEX VECTOR BUNDLES WITH DISCRETE STRUCTURE GROUP

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Every complex vector bundle whose structure group can be reduced to a discrete group, and whose base is a compact polyhedron, becomes trivial after passage to a finite covering of the base.

Theorem. *Let V be a finite-dimensional complex local system of rank n on a compact polyhedron X , and let $\mathcal{V}$ be the corresponding complex vector bundle. There exists a finite surjective covering $\pi : \tilde{X} \rightarrow X$ such that the inverse image of $\mathcal{V}$ on $\tilde{X}$ is trivial as a vector bundle.*

We may assume that X is connected and has a base point p . Once a basis of V_p has been chosen, the local system V corresponds to a homomorphism

$$\rho : \pi_1(X, p) \rightarrow \mathrm{GL}(n, \mathbf{C}).$$

Since $\pi_1(X, p)$ is finitely generated, there is a subring A of $\mathbf{C}$, finitely generated over $\mathbf{Z}$, such that ρ factors through $\mathrm{GL}(n, A)$: choose a finite set of generators (g_i) for $\pi_1(X, p)$ and let A be generated by the matrix coefficients of the g_i . Let m_1 and m_2 be two maximal ideals of A such that the finite fields A/m_1 and A/m_2 have distinct characteristics. We take for $\tilde{X}$ the covering of X corresponding to the subgroup of $\pi_1(X, p)$ consisting of the elements g such that $\rho(g) \equiv 1$ modulo both m_1 and m_2 . Its index divides the order of $\mathrm{GL}(n, A/m_1) \times \mathrm{GL}(n, A/m_2)$. Showing that this $\tilde{X}$ works is equivalent to proving the following proposition.

Proposition. *Suppose that V is defined by a representation $\rho : \pi_1(X, p) \rightarrow \mathrm{GL}(n, A)$, where A is a subring of $\mathbf{C}$ finitely generated over $\mathbf{Z}$. If there are two maximal ideals m_1 and m_2 of A , of distinct residue characteristics, such that ρ is trivial modulo m_1 and modulo m_2 , then $\mathcal{V}$ is trivial.*

Let d and N be integers with $\dim X \leq d$ and N large enough, say $N \geq d/2$. The Grassmannian $G_{n,N}$ of n -planes in $\mathbf{C}^{n+N}$ is a sufficiently good approximation to $B\mathrm{GL}(n, \mathbf{C})$ that

$$[X, G_{n,N}] = [X, B\mathrm{GL}(n, \mathbf{C})],$$

and $\mathcal{V}$ corresponds to a homotopy class of maps $v : X \rightarrow G_{n,N}$. Let X' be the fibration over X whose fiber over $x \in X$ is the space, $(2N+1)$ -connected and hence $(d+1)$ -connected, of linear injections from $\mathcal{V}_x$ into $\mathbf{C}^{n+N}$. Let

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$v' : X' \rightarrow G_{n,N}$ send an injection to its image. The projection $\pi : X' \rightarrow X$ has a section s , unique up to homotopy, and $v = v's$.

$$\begin{array}{ccc} X' & \xrightarrow{v'} & G_{n,N} \\ \uparrow \pi \quad s & \nearrow v & \\ X & & \end{array}$$

The following conditions are equivalent: (a) $\mathcal{V}$ is trivial; (b) $v \sim 0$, i.e. v is homotopic to a constant map; (c) v' is homotopic to zero on the d -skeleton of X' , i.e. for every $w : K \rightarrow X'$ with $\dim K \leq d$ (or just for K the d -skeleton of a triangulation of X'), one has $v'w \sim 0$; (d) the composite

$$v'' : X' \xrightarrow{v'} G_{n,N} \rightarrow \text{cosq}_d(G_{n,N}),$$

where $\text{cosq}_d(G_{n,N})$ is the d -th stage of the Postnikov decomposition of $G_{n,N}$, having the same homotopy groups π_i for $i \leq d$, is homotopic to zero.

The equivalence of (c) and (d) is general for any map v' . The equivalence of (b) and (c) follows from the fact that, for every complex K of dimension $\leq d$, the map $[K, X'] \rightarrow [K, X]$ is an isomorphism.

Since $G_{n,N}$ is simply connected, it follows from the Hasse principle for maps [4, Theorem 3.1] that $v'' \sim 0$ if and only if, for every l , its l -adic completion

$$v''_l : X'^{\wedge}_l \rightarrow \text{cosq}_d(G_{n,N})^{\wedge}_l = \text{cosq}_d((G_{n,N})^{\wedge}_l)$$

is homotopic to zero. The proposition follows from the following lemma.

Lemma. *Suppose V is defined by $\rho : \pi_1(X, p) \rightarrow \text{GL}(n, A)$, where A is a subring of $\mathbf{C}$ finitely generated over $\mathbf{Z}$. If there is a maximal ideal m of A , of residue characteristic different from the prime number l , such that ρ is trivial modulo m , then $v''_l \sim 0$.*

We may suppose that X is a union of faces of the simplex $\Delta \subset \mathbf{R}^M$ spanned by the basis vectors. For each face σ of Δ , let $\sigma_{\mathbf{R}}$ be the affine subspace spanned by σ , and let $\sigma_{\mathbf{Z}}$ be the affine subvariety of affine space $\mathbf{A}_{\mathbf{Z}}^M$ over $\mathbf{Z}$ whose real points are $\sigma_{\mathbf{R}}$. Let $X_{\mathbf{Z}}$ be the union of the $\sigma_{\mathbf{Z}}$ for $\sigma \subset X$. This is a scheme over $\mathbf{Z}$; for any ring B , denote by X_B the scheme over B obtained from it by extension of scalars. The inclusion of X into $X_{\mathbf{Z}}(\mathbf{C}) \subset \mathbf{C}^M$ is a homotopy equivalence. For $l \neq p$, the mod l cohomology of X , or of $X_{\mathbf{Z}}(\mathbf{C})$, agrees with the mod l etale cohomology of $X_{\overline{\mathbf{F}}_p}$. By Mayer-Vietoris, this follows from the corresponding fact for affine space (SGA 4, XV 2.2).

The representation ρ gives a vector bundle $\mathcal{V}_A$ on X_A . The bundle $\mathcal{V}$ is obtained by extension of scalars to $\mathbf{C}$, and its reduction modulo m is trivial. The definition of X' is purely algebraic, and gives a diagram of schemes over A :

$$X_A \xleftarrow{s_A} X'_A \xrightarrow{v'_A} (G_{n,N})_A.$$

Use a subscript 1 for the analogous objects attached to the unit representation of $\pi_1(X, p)$ in $\mathrm{GL}(n, A)$. The morphisms of schemes v'_A and v'_{1A} become isomorphic after reduction modulo m . If k is an algebraic closure of A/m , and if B is the strict Henselization of A at k , then there is a chain of isomorphisms between morphisms of l -completed homotopy types:

$$v'_l \sim (v'_C)_l \sim (v'_B)_l \sim (v'_k)_l \sim (v'_{1k})_l \sim (v'_{1B})_l \sim (v'_{1C})_l \sim v'_{1l}.$$

Therefore

$$v''_l \sim v''_{1l} \sim 0.$$

In this chain, the two extreme terms are completions of morphisms between ordinary spaces [4, Section 3]; the other terms are completions of morphisms between étale homotopy types [1].

Remarks. (a) If N is large and $2p \geq d$, the map induced by Chern classes

$$\mathrm{cosq}_d(G_{n,N}) \rightarrow \mathrm{cosq}_d\left(\prod_1^n K(\mathbf{Z}, 2i)\right)$$

is an isomorphism after localization at p . One may deduce that the hypothesis of the proposition can be replaced by the following one: ρ is trivial modulo a maximal ideal m of residue characteristic p such that $H_*(X)$ has no p -torsion and $2p \geq \dim X$.

(b) In (a), and in the proposition, one may replace “ ρ is trivial modulo m ” by “modulo m , the image of ρ is contained in the unipotent radical of a Borel subgroup.”

Remarks. (c) The theorem strengthens the differential-geometric theorem [3] asserting that the real Chern classes of a vector bundle with integrable connection are trivial. Its proof recalls Grothendieck’s algebraic proof of that fact [2].

(d) The problem was suggested to one of us by [5]; the theorem extends the range of application of computations made there (Theorems C.2 and D.2).

(e) The analogue of the theorem for real vector bundles with discrete structure group in $\mathrm{SL}(n, \mathbf{R})$ is false: the real Euler class may be nonzero already for $n = 2$ and X a Riemann surface.

Problem. Does the real analogue of the theorem hold for bundles whose structure group can be reduced to $\mathrm{SL}(n, \mathbf{Z})$ [6], or for n odd? Is the real Euler class the only obstruction?

Remarks. (f) The technique of reducing modulo two maximal ideals, and a problem of Atiyah about vector bundles, are described in the introduction to [1]. The algebraic classifying map, namely the diagram above, and the methods of [5] used here, solve Atiyah’s problem.

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